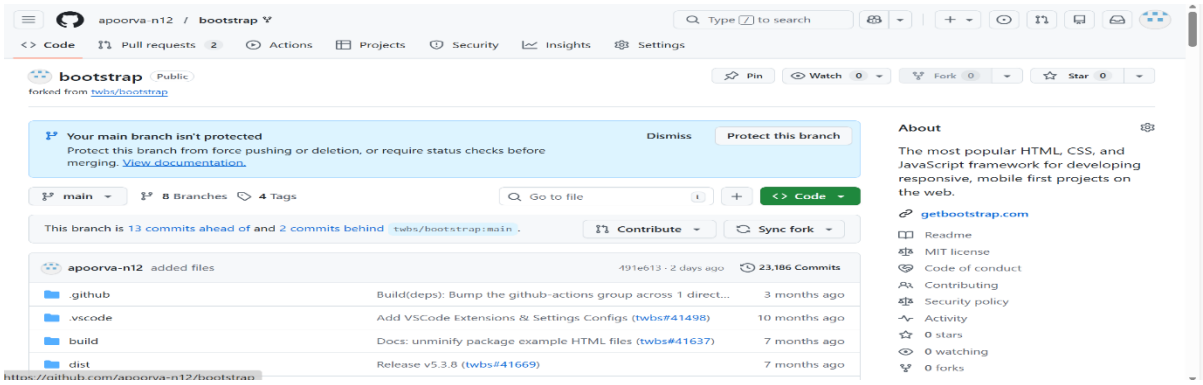
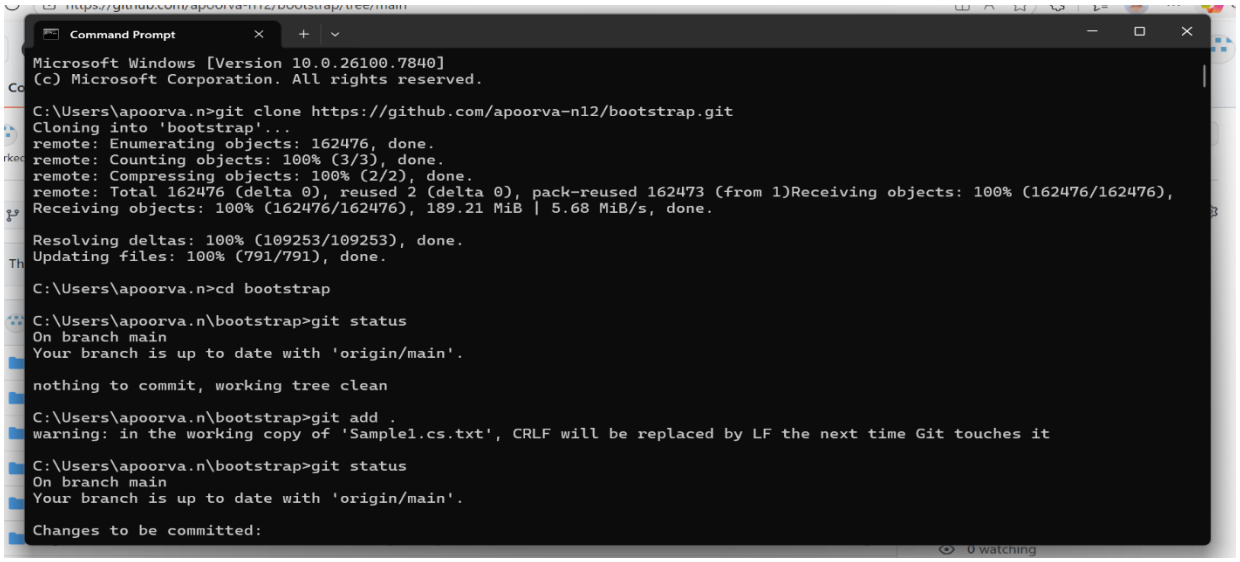
Git Assessment:

Git commands:

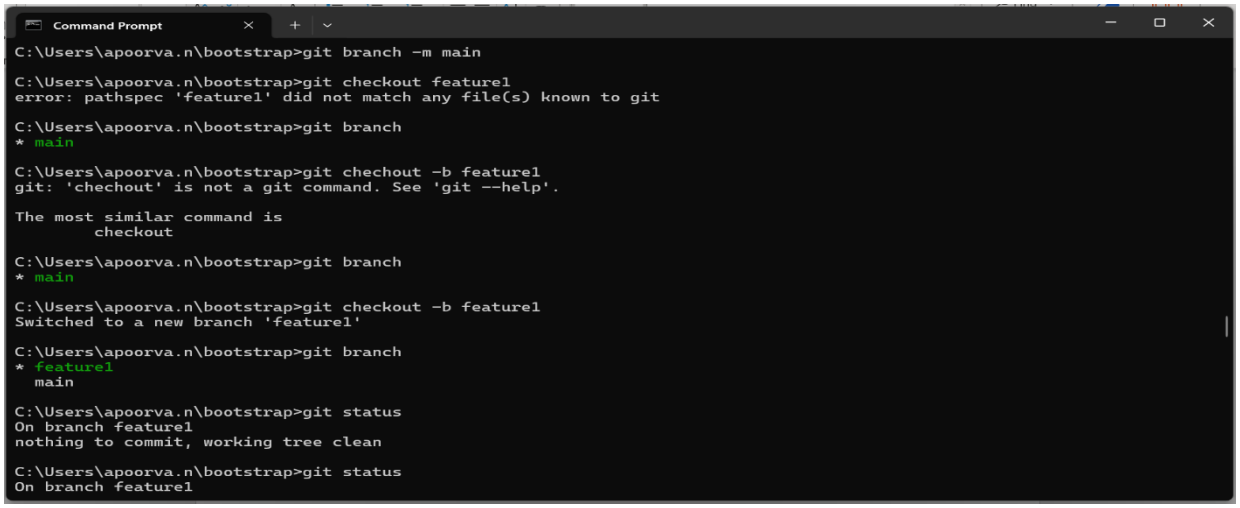
git fork: is a copy of someone else's repository that lives under your own account.



git clone: is a command used to **download a repository from a remote server**



Branch Creation and check out to different branch: Branch creation in Git means creating a **separate line of development** from the main codebase. This lets you work on features or fixes **without affecting the main branch**.



git add . : Stage all changed files.

```
Command Prompt
Microsoft Windows [Version 10.0.26100.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\apoorva.n>git clone https://github.com/apoorva-n12/bootstrap.git
Cloning into 'bootstrap'...
remote: Enumerating objects: 162476, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 162476 (delta 0), reused 2 (delta 0), pack-reused 162473 (from 1)Receiving objects: 100% (162476/162476),
Receiving objects: 100% (162476/162476), 189.21 MiB | 5.68 MiB/s, done.

Resolving deltas: 100% (109253/109253), done.
Updating files: 100% (791/791), done.

C:\Users\apoorva.n>cd bootstrap

C:\Users\apoorva.n\bootstrap>git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

C:\Users\apoorva.n\bootstrap>git add .
warning: in the working copy of 'Sample1.cs.txt', CRLF will be replaced by LF the next time Git touches it

C:\Users\apoorva.n\bootstrap>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
```

git commit: It records **changes to tracked files.**

```
Command Prompt

C:\Users\apoorva.n\bootstrap>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   Sample1.cs.txt


C:\Users\apoorva.n\bootstrap>git commit -m "Added new Sample1.txt"
[main 142c936d6] Added new Sample1.txt
 1 file changed, 17 insertions(+)
 create mode 100644 Sample1.cs.txt

C:\Users\apoorva.n\bootstrap>git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 18 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 449 bytes | 89.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/apoorva-n12/bootstrap.git
   fbde78520..142c936d6  main -> main

C:\Users\apoorva.n\bootstrap>git branch -m main

C:\Users\apoorva.n\bootstrap>git checkout feature1
error: pathspec 'feature1' did not match any file(s) known to git
```

git pull: used to fetch the latest changes from the remote repository and merge them into the local branch.

```
MINGW64:/c/Users/apoorva.n/bootstrap

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch2)
$ git push origin branch2
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 18 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 528 bytes | 29.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/apoorva-n12/bootstrap.git
   09f12ea50..a74295d61  branch2 -> branch2

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch2)
$ git pull origin branch2
From https://github.com/apoorva-n12/bootstrap
 * branch                branch2     -> FETCH_HEAD
Already up to date.

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch2)
$ ^C

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch2)
$ |
```

git stash: temporarily save the uncommitted changes.

```
Command Prompt
C:\Users\apoorva.n\bootstrap>git switch branch2
Switched to branch 'branch2'

C:\Users\apoorva.n\bootstrap>git swich main
git: 'swich' is not a git command. See 'git --help'.

The most similar command is
        switch

C:\Users\apoorva.n\bootstrap>git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

C:\Users\apoorva.n\bootstrap>git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Sample stash.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\apoorva.n\bootstrap>git stash save "WIP: editing test.txt"
No local changes to save

C:\Users\apoorva.n\bootstrap>git stash push -u -m "WIP: editing test.txt"
Saved working directory and index state On main: WIP: editing test.txt

C:\Users\apoorva.n\bootstrap>
```

```
Command Prompt
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Sample stash.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\apoorva.n\bootstrap>git stash save "WIP: editing test.txt"
No local changes to save

C:\Users\apoorva.n\bootstrap>git stash push -u -m "WIP: editing test.txt"
Saved working directory and index state On main: WIP: editing test.txt

C:\Users\apoorva.n\bootstrap>git stash list
stash@{0}: On main: WIP: editing test.txt

C:\Users\apoorva.n\bootstrap>git stash apply
Already up to date.
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Sample stash.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\apoorva.n\bootstrap>
```

git merging:

three way merge: integrates changes from another branch into the current branch.

```
MINGW64:/c/Users/apoorva.n/bootstrap
INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$ git checkout -b hotfix
Switched to a new branch 'hotfix'

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (hotfix)
$ git status
On branch hotfix
nothing to commit, working tree clean

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (hotfix)
$ git status
On branch hotfix
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Sample5.txt

nothing added to commit but untracked files present (use "git add" to track)

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (hotfix)
$ git add .

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (hotfix)
$ git commit -m "added hotfix branch"
[hotfix 5b7a292ae] added hotfix branch
 1 file changed, 1 insertion(+)
 create mode 100644 Sample5.txt

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (hotfix)
$ git push origin hotfix
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 18 threads
```

```
MINGW64:/c/Users/apoorva.n/bootstrap
INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (hotfix)
$ git push origin hotfix
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 18 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 288 bytes | 41.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'hotfix' on GitHub by visiting:
remote:      https://github.com/apoorva-n12/bootstrap/pull/new/hotfix
remote:
To https://github.com/apoorva-n12/bootstrap.git
 * [new branch]          hotfix -> hotfix

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (hotfix)
$ git switch main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$ git merge hotfix
Updating 8ad1e706b..5b7a292ae
Fast-forward
 Sample5.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 Sample5.txt

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$ ^C
```

```
MINGW64:/c/Users/apoorva.n/bootstrap
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'hotfix' on GitHub by visiting:
remote:      https://github.com/apoorva-n12/bootstrap/pull/new/hotfix
remote:
To https://github.com/apoorva-n12/bootstrap.git
 * [new branch]          hotfix -> hotfix

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (hotfix)
$ git switch main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$ git merge hotfix
Updating 8ad1e706b..5b7a292ae
Fast-forward
 Sample5.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 Sample5.txt

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$ ^C

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/apoorva-n12/bootstrap.git
   32671d58e..5b7a292ae  main -> main

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$
```

Fast Forword merge: Main branch has not changed since you created another branch.

```
MINGW64:/c/Users/apoorva.n/bootstrap
INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch2)
$ git switch main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 35 commits.
  (use "git push" to publish your local commits)

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$ git merge branch2
Merge made by the 'ort' strategy.
 fast_forward.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 fast_forward.txt

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$ ^C

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$ fit log --oneline --graph
bash: fit: command not found

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$ git log --oneline --graph
*   50bdcb615 (HEAD -> main) Merge branch 'branch2'
```

Merge conflict: Git happens when **two branches change the same part of a file**, and Git **cannot automatically decide which change to keep**.

```
MINGW64:/c/Users/apoorva.n/bootstrap
INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main|MERGING)
$ git conflict --global merge.tool vscode
git: 'conflict' is not a git command. See 'git --help'.

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main|MERGING)
$ git config --global merge.tool vscode

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main|MERGING)
$ git mergetool
git config option merge.tool set to unknown tool: vscode
Resetting to default...

This message is displayed because 'merge.tool' is not configured.
See 'git mergetool --tool-help' or 'git help config' for more detail
'git mergetool' will now attempt to use one of the following tools:
opendiff kdiff3 tkdiff xxdiff meld tortoisemerge gvimdiff diffuse di
rge p4merge araxis bc codecompare smerge emerge vimdiff nvimdiff
Merging:
merge_conflicts.txt

Normal merge conflict for 'merge_conflicts.txt':
  {local}: modified file
  {remote}: modified file
Hit return to start merge resolution tool (vimdiff):
```

git cherry-pick: commit copy between branches (local).

```
MINGW64:/c/Users/apoorva.n/bootstrap
c9e83a12b Build(deps-dev): Bump vnu-jar from 25.11.8 to 25.11.17

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch4)
$ git cherry_pick 2f4996004
git: 'cherry_pick' is not a git command. See 'git --help'.

The most similar command is
        cherry-pick

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch4)
$ git cherry-pick 2f4996004
[branch4 a61325bfb] added text of cherry_pick2
 Date: Sun Mar 15 20:44:32 2026 +0530
 1 file changed, 1 insertion(+)
 create mode 100644 cherry_pick2.txt

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch4)
$ git log --oneline --graph
* a61325bfb (HEAD -> branch4) added text of cherry_pick2
* 96e72e0b7 added text of cherry pick
* b27da0e81 added text of amend1
* 4bae9de19 updated text commit
* 0ab64c9a2 first commit file
* 3f1d0780d restore added text
```

rebasing: means moving your branch commits on top of another branch to maintain a clean commit history.

```
MINGW64:/c/Users/apoorva.n/bootstrap
INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$ git commit -m "added text of rebase1"
[main eff489f82] added text of rebase1
 1 file changed, 2 insertions(+), 1 deletion(-)

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$ git switch branch2
Switched to branch 'branch2'

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch2)
$ git rebase main
Successfully rebased and updated refs/heads/branch2.

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch2)
$ git log --oneline
77a5ccbea (HEAD -> branch2) added text of rebase
eff489f82 (main) added text of rebase1
7b98ff766 added text of rebase
7e423362e added text to rebase
5e032f4f2 Merge branch 'branch2'
d17c2c59b merge_conflicts1
436229791 merge conflicts
8128bd483 merge conflict
9b65f07f7 commit merge conflix
```

git tag: marking a specific commit with a label (tag) so it's easy to find later.

```
MINGW64:/c/Users/apoorva.n/bootstrap

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch2)
$ git tag v1.0
fatal: tag 'v1.0' already exists

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch2)
$ git tag v1.0

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch2)
$ git tag -a v1.0 -m "version of text"
fatal: tag 'v1.0' already exists

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch2)
$ git tag
v1.0
v1.0
v2.0
v3.0
v4.0

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (branch2)
$ git show v1.0
commit bd206a77486e3869b926994294a0aca04909e80b (HEAD -> branch2, tag: v1.0)
Author: Apoorva <apoorva.n@intimetec.com>
```

git revert: It **creates a new commit that undoes the changes** introduced by a previous commit.

```
MINGW64:/c/Users/apoorva.n/bootstrap
fd912946a initial commit

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$ git revert HEAD
[main d7da41e60] Revert "added text of revert2"
 1 file changed, 1 insertion(+), 1 deletion(-)

INTIMETEC+apoorva.n@ITT-APOORVA-N MINGW64 ~/bootstrap (main)
$ git log --oneline
d7da41e60 (HEAD -> main) Revert "added text of revert2"
244515c77 added text of revert1
1a7d940f5 added text of revert
eff489f82 (origin/main, origin/HEAD) added text of rebase1
7b98ff766 added text of rebase
7e423362e added text to rebase
5e032f4f2 Merge branch 'branch2'
d17c2c59b merge_conflicts1
436229791 merge conflicts
8128bd483 merge conflict
9b65f07f7 commit merge conflix
46479642f merge conflict
75604ce76 added text of merge conficts1
ec4a84a12 add text of merge conflicts
50bdcb615 Merge branch 'branch2'
```